

SIMS — Multi-School Test Scenarios (3 schools)

Purpose: exercise the full platform end-to-end across three schools — SBA engine, teacher identity & scoping, cross-school transfers, reports, transcripts — with realistic data. **Build under test:** commit f013d02 (Phase 8 core complete). **Date:** 2026-07-06.

You provision the schools and create the students/teachers; this doc says exactly what to enter and what to verify. Work through it in order — later scenarios depend on earlier setup.

Accounts & roles — read first

- **Provision schools** as a **super_admin** (schoolbasedassessment1@gmail.com).
- Each provisioned school gets **one school_admin** (temp password shown once — copy it).
- **Teacher** logins are created from **Teachers** → **profile** → **Create login account** (role teacher).
- **There is no UI to create hod / head_teacher accounts yet.** school_admin is allowed at every SBA step (moderate + approve), so **the school_admin plays HOD and Head Teacher**. Use a real **teacher** account only to verify teacher scoping.
- Tip: use two browsers/profiles (or one normal + one incognito) so you can be signed in as an admin and a teacher at once.

Part 0 — Provision the three schools (as super_admin)

Register each via **Register School**. Record the generated **SIMS code** (SCH-0000xx) — you'll need the receiving school's code for transfers.

#	Name	EMIS (you enter)	Type	Ownership	Province	District	SIMS code (record)
A	Discovery Secondary School	10001	Secondary	Government	Lusaka	Lusaka	SCH- _____
B	Kansenshi Secondary School	20002	Secondary	Government	Copperbelt	Ndola	SCH- _____
C	Kasama Boys Secondary School	30003	Secondary	Government	Northern	Kasama	SCH- _____

- ☐ All three provisioned; three admin temp passwords copied; three SIMS codes recorded.
 - ☐ For each school, sign in as its admin once (this backfills claims), open **School Profile**, confirm the EMIS shows.
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Part 1 — Per-school setup (as each `school_admin`)

Do this for **School A** fully first (Scenarios 1–3 run on A), then a lighter setup for **B** and **C** (they mainly receive transfers).

For **each** school, signed in as its admin:

1. **Academic context** — in the header bar, pick the **Academic Year** and **Term** (e.g. AY-2026 · Term 1). Confirm Form 2 / Form 3 exist under Academic Structure.
2. **Streams** — Academic Structure → Streams: create **F2-A** and **F3-A** (capacity 40).
3. **Subjects** — Subjects: create three, **SBA enabled**, forms offered **F2, F3**:
 - MATH Mathematics · ENG English Language · BIO Biology
4. **Teachers** — register two, then **Create login account** for each (copy temp passwords):
 - School A: **Mr John Banda** (Maths dept), **Mrs Grace Phiri** (Languages)
 - School B: **Mr Peter Mwale**, **Mrs Sarah Zulu**
 - School C: **Mr David Tembo**, **Mrs Mary Sakala**
5. **Teaching assignments** — Teaching (pick Year+Term in header): assign
 - teacher 1 → **MATH** → **F2-A** (confirm the form shows the teacher's **Department**)
 - teacher 2 → **ENG** → **F2-A**
6. **Students** — Admit **6 learners into F2-A** (Admission wizard; admission number is auto, EMIS auto from the school). Suggested names so transfers are traceable:
 - School A: Liam O'Brien, Emma Sullivan, Noah Kelly, Olivia Kennedy, Mason Byrne, Ava Ryan
 - School B: Ethan Burke, Mia Fitzpatrick, Jackson Murphy, Charlotte O'Donnell
 - School C: Aiden Hayes, Amelia Brennan, Grayson Foley, Harper Doyle
 - ☐ After admitting, open a student → confirm a **SIMS Learner ID** (SL-2026-...) appears (it's minted by a function ~1–2 s after admission; refresh if not shown).

Faster alternative for A: as the admin, use **SBA Demo Data** (/dev/seed) to seed a Form 2 class in one click, and/or **Bulk Admit** for many learners. But provision B and C's admins/streams/subjects manually (the seeders only touch the signed-in school).

Scenario 1 — SBA happy path (School A)

Signed in as **School A admin** unless noted.

1. **SBA Plans** → New Plan → Form 2 → Mathematics → add 4 tasks (e.g. Test 1 /20, Test 2 /20, Assignment /20, Project /20) → **Save** → **Publish**.
2. Sign in as **Mr Banda (teacher)** in the other browser. He lands on **My Classes** → he sees **Mathematics · F2-A** with the learner count and "Marks entry".
 - ☐ Expected: only *his* class(es) show — no dropdowns.
3. Mr Banda opens the class → enters marks for all 6 learners → **Save** → **reload the browser** → marks persisted → **Submit for moderation**.
 - ☐ Submit is blocked until every learner is fully scored + saved.
4. Back as **admin** → **SBA Review** → "Needs review" shows the class → **Moderate** → then **Approve** (add a note if you like).
5. Wait ~3 s (freeze function) → reopen the class in **SBA Marks** → it's **locked/read-only**.
 - ☐ A learner's mark can no longer be edited.
6. Verify results:
 - ☐ **Student profile** → **SBA tab**: Form 2 raw %, band, **frozen · approved**.
 - ☐ **SBA Readiness**: Mathematics · F2 shows **Ready**.
 - ☐ **Reports**: KPIs update; Compliance-by-subject bar for Mathematics moves toward Ready.
 - ☐ **Register** (click the class chip on Readiness): printable list.

Scenario 2 — Return, resubmit, versioning (School A)

1. Repeat a plan+marks for **English (ENG) F2-A** as **Mrs Phiri**, and **Submit**.
2. As **admin** → **SBA Review** → **Return** with a note ("recheck Test 2").
3. As **Mrs Phiri** → the class is editable again → change a mark → **Save** → **Submit**.
 - ☐ On Review the status badge shows **v2**; **History** lists submitted → returned (with your note) → submitted.
4. As **admin** → Moderate → Approve.

Scenario 3 — Teacher scoping (School A) ☐ security

1. As **Mr Banda (teacher)**, open **SBA Marks** and try to select **English · F2-A** (Mrs Phiri's class).
 - ☐ Expected: the grid loads **read-only** (he isn't the assigned teacher) — he can't Save.
 - Note: read-scoping isn't implemented yet, so he can still *view* it; the check is that he **cannot save**.
2. Confirm Mr Banda has **no Moderate/Approve** actions anywhere (only admin/head/hod do).

Scenario 4 — Cross-school transfer A → B □ the big one

1. As **School A admin**, open **Liam O'Brien's profile** → confirm his **SIMS Learner ID**.
2. Click **Transfer out** → Receiving school code = **School B's SIMS code** → Reason "Parent relocation" → date → **Send transfer request**.
 - Success message; on School A's **Transfers** → **Outgoing**, the request shows **requested**.
3. Sign in as **School B admin** → **Transfers** → **Incoming** → the request from Discovery appears.
 - **View record** shows Liam's identity, enrollment history, and any SBA results (the "digital envelope").
4. Click **Accept** → wait ~5 s → **refresh**.
 - The request moves **accepted** → **completed** ("learner imported").
5. Verify the move:
 - **School B** → **Students registry**: **Liam O'Brien** now exists, **same SIMS Learner ID**, status active.
 - **School A** → **Liam's profile**: status is now **transferred**; his transcript/history still shows his School A record (history preserved).
 - **Known limitation**: at School B, Liam has **no stream yet** (import leaves placement to you) — he's in the registry but not in a class roster. There's no placement screen yet; this is expected.

Scenario 5 — Transfer chain B → C (identity continuity)

1. As **School B admin**, open **Liam O'Brien** → **Transfer out** → Receiving code = **School C's SIMS code** → send.
2. As **School C admin** → **Transfers** → **Incoming** → **Accept** → refresh.
 - Liam appears at **School C** with the **same SIMS Learner ID** as at A and B — the identity is continuous across all three schools.
 - School B's Liam is now **transferred**; School A's remains **transferred**. Three enrollment records, one learner.

Scenario 6 — Exam numbers, transcript, export (School A)

1. **Examination Numbers** (/assessments/exam-numbers) → Form 2 → F2-A → enter numbers for the learners → **Save**.
2. Open a learner → **Transcript** → **Print / Save as PDF**.
 - Clean sheet (no app sidebar/header): school + EMIS, learner identity + **SIMS Learner ID**, enrollment history, SBA results, signature line.
3. **ECZ Export** (/assessments/export) → Form 2 → Mathematics → tiles (Ready / Missing exam# / Not approved) → **Export CSV**.
 - CSV opens with raw marks keyed by exam number; a name like =x is **not** executed (injection guard).

Scenario 7 — Role & security spot-checks

- ☐ A **teacher** account has **no** School Profile / Reports / SBA Review / Transfers nav.
 - ☐ A **teacher** cannot open another school's data at all (multi-tenant): they only ever see their own school.
 - ☐ On **Transfers**, a teacher has no "Transfer out" button (initiating is admin/head only).
 - ☐ School B cannot see School A's transfer requests **except** the one addressed to School B.
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Known limitations to expect (not bugs)

- **Transferred learner has no stream** at the receiving school until placed — no placement UI yet (follow-up 8.3b).
- **hod/head_teacher** role accounts can't be created in the UI — admin plays those roles.
- **Read-scoping** of SBA sheets isn't done — a teacher can *view* (not save) another class's sheet.
- **Existing/older students** (admitted before the Learner-ID feature) have **no** SIMS Learner ID — use freshly admitted learners for transfer tests.
- **Guardians/medical/documents** aren't in the transfer envelope yet (identity + enrollment + SBA only).
- **Offline:** the *first* save of a brand-new class needs connectivity; later saves work offline.
- Learner ID and SBA freezing are done by Cloud Functions — allow a few seconds + refresh.

Pass criteria → tag v1.0.0-beta

Scenarios 1–6 pass with realistic data, no console errors, marks immutable after approval, the A→B→C transfer preserves one Learner ID, and the transcript/export come out clean.